

Mid-Point Check

Group 02 | Colby Valcourt and Suleman Osman

Project Proposal:

Chosen Approach: Option 1: Time-Series Forecasting + Regression-Based OPF.

Literature Review:

Based on the literature review performed on [1] & [2] XGBoost is trained on historical weather data, application of feature engineering to identify influential predictors and finally cross validation. XGBoost proved to be more accurate in time series forecasting compared to methods like SVR in [3] and [4] due to SVRs inaccuracy when handling large amounts of data. Particle Swarm Optimization (PSO) is mentioned in [4] as a useful tool to determine the optimal dispatch schedule for the ESS by optimizing the hyperparameters in an efficient manner. PSO ensures faster convergence and improved accuracy by avoiding local minima during training and, when paired with something like XGBoost, will allow for quick and accurate dispatch schedule creation for the ESS.

Proposed Methodology:

Stage 1 – Load Forecasting

In Stage 1 we will develop 6 hours ahead load forecasting model using extreme gradient boosting (XGBoost). The historical load data will be transformed into a supervised learning problem by creating time-based features such as hour, day of week, quarter, and temperature will be included to capture weather-driven variations. The forecast (output) will be the input for Stage 2 Regression-Based Optimal Power Flow (OPF).

Stage 2 – Regression-Based Optimal Power Flow (OPF)

In Stage 2 we will design an OPF solver using PSO to determine and train the model on optimal hyperparameters before using XGBoost again to create a dispatch schedule in accordance with the supervised learning forecast schedule.

Preliminary Implementation Plan (Timeline):

We plan on following the recommended timeline provided in the project handout. The only planned modification is that we will start working on the core model on March 9th rather than March 13th as the handout recommends.

Division of Responsibilities:

We plan on working together in all areas of the project. Suleman is going to be working on stage 1 primarily while Colby will work more on stage 2. All other tasks will be performed together in a 50/50 workload split.

Git Repository Link: <https://github.com/VColby/ENEL-619-Final-Project>

Sources:

- [1] - XGBoost-Based Day-Ahead Load Forecasting Algorithm Considering Behind-the-Meter Solar PV Generation
- [2] - Wavelet-Enhanced Hybrid LSTM-XGBoost Model for Predicting Time Series Containing Unpredictable Events
- [3] - Support Vector Regression Approach to Optimal Power Flow Analysis in Smart Grids with IoT
- [4] - Intelligent Synergy Leveraging Artificial Intelligence for Renewable Energy Forecasting and Grid Optimization